

MONTE REAL AB, Portugal

WMO: 085400

Lat: 39.828N

Lon: 8.883W

Elev: 177

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	35.6	28.0	22.1	40.7	30.6	24.9	41.2	23.2	52.5	20.7	53.2	2.9	160	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.1	88.4	67.2	84.1	66.2	79.8	65.3	70.1	80.6	68.8	77.9	67.6	75.6	9.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	100.2	72.4	66.1	97.0	71.3	64.8	92.3	70.3	34.1	80.4	32.9	77.6	32.1	75.9	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.5	16.4	DB	28.8	96.3	2.2	3.9	27.2	99.1	25.8	101.4	24.6	103.7	23.0	106.5	(4)
(5)			WB	28.1	73.3	2.4	2.8	26.3	75.3	24.9	77.0	23.6	78.6	21.8	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.7	50.8	51.5	54.9	58.0	61.5	65.3	68.0	68.7	67.1	63.6	55.1	51.8		
	DBStd	7.96	4.63	4.59	4.57	4.74	4.84	4.09	3.59	3.68	3.90	5.00	5.47	5.10		
	HDD50	151	48	34	10	1	0	0	0	0	0	0	16	42		
	HDD65	2345	442	378	313	216	134	41	5	3	21	84	298	410		
	CDD50	3705	71	76	163	240	357	460	557	580	513	423	169	96		
	CDD65	428	0	0	1	6	26	51	97	119	84	42	2	0		
	CDH74	2686	0	0	29	86	257	349	501	707	474	276	7	0		
	CDH80	890	0	0	3	15	78	119	203	263	140	69	0	0		
(14) Wind	WSAvg	7.2	6.6	7.1	7.5	7.5	7.7	7.9	8.4	7.9	6.3	6.4	6.3	6.7		
(15) Precipitation	PrecAvg	30.7	4.5	3.2	2.5	2.8	2.4	0.7	0.3	0.5	1.6	4.3	4.4	3.9		
	PrecMax	44.5	11.7	7.7	7.6	8.6	5.6	2.4	1.4	1.7	4.4	9.2	15.4	12.5		
	PrecMin	16.9	0.3	0.4	0.3	1.1	0.0	0.0	0.0	0.0	0.2	0.4	0.5	0.2		
	PrecStd	7.0	2.9	2.1	2.0	1.7	1.5	0.7	0.4	0.5	1.3	2.2	3.2	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.7	69.5	77.3	82.4	87.8	90.0	94.1	94.4	89.6	86.5	74.4	66.3		
		MCWB	57.0	56.2	58.3	64.3	66.3	68.1	70.4	71.4	66.5	64.4	61.8	57.1		
	2%	DB	61.9	65.2	72.1	75.7	81.0	82.7	85.8	87.5	84.1	80.7	69.5	63.7		
		MCWB	56.1	54.8	57.1	60.4	64.1	66.0	67.2	68.1	66.6	64.4	60.3	57.6		
	5%	DB	60.3	62.2	67.9	71.4	75.3	77.2	78.3	81.0	78.9	76.6	66.2	61.6		
		MCWB	55.9	54.5	56.1	59.6	62.5	65.3	67.0	66.9	65.9	63.4	60.3	56.7		
	10%	DB	58.7	59.5	63.2	66.6	70.4	73.3	74.9	76.7	75.3	72.5	64.1	60.1		
		MCWB	55.3	54.4	55.6	58.4	60.7	63.9	65.7	66.5	65.4	63.2	59.0	55.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	59.4	62.4	65.7	68.4	70.0	72.9	73.3	70.2	69.8	66.3	61.2		
		MCDB	61.9	62.3	71.0	77.0	82.0	83.1	88.8	87.9	79.7	77.7	68.7	63.1		
	2%	WB	58.5	57.9	59.8	62.6	65.9	67.9	69.5	70.1	68.6	67.9	63.3	59.7		
		MCDB	60.3	61.3	66.5	72.2	76.9	78.5	79.5	80.3	77.5	74.0	66.6	61.8		
	5%	WB	57.3	56.8	58.1	60.5	63.5	66.2	67.9	68.6	67.3	66.5	61.5	58.1		
		MCDB	59.1	60.4	64.1	68.5	72.3	75.8	75.5	77.1	75.4	71.9	64.7	60.7		
	10%	WB	55.9	55.2	56.6	58.9	61.6	64.7	66.5	67.4	66.2	64.9	59.8	56.4		
		MCDB	57.9	59.2	62.1	65.3	68.6	71.7	73.0	74.6	73.2	70.4	63.1	59.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	15.8	15.9	15.4	15.9	15.2	15.1	17.1	17.7	17.7	15.9	15.2		
		MCDBR	16.5	20.4	25.5	24.9	25.9	24.6	24.4	27.4	25.4	25.2	18.9	15.6		
	5% WB	MCWBR	11.4	12.1	13.0	12.3	12.0	10.7	10.3	11.4	10.8	11.5	11.3	10.5		
		MCWBR	10.6	14.1	19.0	20.3	21.4	20.1	19.1	20.3	18.8	17.5	12.6	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.345	0.377	0.375	0.382	0.387	0.375	0.388	0.383	0.369	0.342	0.330		
	taud		2.496	2.453	2.368	2.386	2.394	2.399	2.423	2.391	2.424	2.472	2.503	2.500		
	Ebn at Noon		264	277	278	285	284	282	285	278	272	264	258	255		
	Edh at Noon		25	29	35	37	37	37	36	36	33	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		657	975	1319	1714	2066	2243	2295	2089	1623	1090	733	592		
	RadStd		95	124	161	175	175	132	104	87	118	102	106	76		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page